



3D Printing Software Developer

QP Code: SSC/Q8902

Version: 3.0

NSQF Level: 5

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Qualification Pack

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SSC/Q8902: 3D Printing Software Developer

Brief Job Description

The job roles emphasize a comprehensive approach to software development within the 3D printing industry, featuring requirements analysis, technical design, programming, and lifecycle management in addition to essential employability skills. Professionals in these roles are expected to collaborate with various stakeholders, possess strong analytical and technical skills, and adapt to the rapidly evolving technology landscape in additive manufacturing.

Personal Attributes

This job requires the individual to have thorough knowledge of various technology trends and processes as well as have updated knowledge about IT initiatives. He/she should be highly motivated and energetic with the ability to self-direct daily activities.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8907: Determine the functional, non-functional, and infrastructural requirements for the defined scope of the software solution](#)
2. [SSC/N8908: Translate the 3dp processes into relevant application storyboards, technical wireframes, and data flows in the software architecture](#)
3. [SSC/N8909: Build high-performance software applications by developing embedded code, data structures and algorithms](#)
4. [SSC/N8910: Monitor, maintain and manage applications throughout the software development lifecycle](#)
5. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	3D Printing
Country	India

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NSQF Level	5
Credits	18
Aligned to NCO/ISCO/ISIC Code	NCO- 2015/2512
Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (PG/UG/Degree/Diploma in (Engg./Science)) OR Completed 3 year diploma after 10th (PG/UG/Degree/Diploma in (Engg./Science)) with 1.5 years of experience in relevant field. Relevant experience in job roles related to 3D Printing/ Additive Manufacturing The relevant experience would include work, internship and apprenticeship undertaken post completion of relevant educational qualification. OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience in relevant field. Relevant experience in job roles related to 3D Printing/ Additive Manufacturing The relevant experience would include work, internship and apprenticeship undertaken post completion of relevant educational qualification
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NIL
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQF Approval Date	07/10/2025
Version	3.0
Reference code on NQR	QG-05-IT-03662-2025-V2-NASSCOM
NQR Version	2

Remarks:

NA

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SSC/N8907: Determine the functional, non-functional, and infrastructural requirements for the defined scope of the software solution

Description

This unit is about analysing the business process outcomes and mapping all the technical requirements which specify the behaviour and quality attributes of the software application

Scope

The scope covers the following :

- Requirements Elicitation and Analysis, Functional requirements, Non-functional requirements, Data Modelling, Resource Identification, Infrastructure Design, System Integration, Technology Integration, Infrastructure Management, Compliance and security, Scalability and Capacity Planning, Service Level Agreements (SLAs)

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Collaborate effectively with cross-functional teams (designers, engineers, manufacturing, marketing) to elicit and document comprehensive requirements. This includes understanding user needs, business goals, and technical constraints. Methods employed should include interviews, workshops, surveys, and document analysis.
- PC2.** Create a detailed requirements specification document covering both functional and non-functional requirements, this should include use cases, user stories, functional requirements and non-functional requirements
- PC3.** Create detailed description of the application's features and capabilities (e.g., slicing algorithms, printer control, material management, error handling, post-processing support, data visualization, reporting, information sharing, etc.).
- PC4.** Create specifications related to performance (speed, scalability, throughput), security (authentication, authorization, data protection), reliability, usability, maintainability, and compliance (regulatory standards).
- PC5.** Define the types and structures of data to be processed and stored by the application. This includes designing the database schema and considering data integrity, consistency, and security.
- PC6.** Identify and document all physical and virtual resources required for development, deployment, and operation of the 3D printing application. This includes hardware (servers, printers, network equipment), software (operating systems, databases, middleware), and personnel
- PC7.** Define the application infrastructure (architecture, deployment environment, network topology) necessary to support the 3D printing process. Address considerations such as high availability, scalability, and security
- PC8.** Determine how the application will integrate with existing systems (e.g., CAD software, ERP systems, cloud storage, other software applications). Detail the methods of integration (APIs, data exchange protocols, etc.) and address any interoperability challenges

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- PC9.** Describe how the application will leverage other technologies such as AI/ML (for optimization and predictive maintenance), cloud computing (for scalability and cost-efficiency), IoT (for real-time monitoring and control), and other relevant technologies.
- PC10.** Identify and document the tools and processes needed for efficient infrastructure management (configuration management, monitoring, logging, security management, automated deployment).
- PC11.** Define security requirements and strategies to protect sensitive data and ensure compliance with relevant regulations (e.g., GDPR, HIPAA).
- PC12.** Define the scalability and capacity requirements of the application to ensure it can handle projected growth in data volume, user base, and processing demands. Document capacity planning methods used
- PC13.** Define appropriate SLAs for the 3D printing application, specifying performance targets, availability, and support

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Proficiency in various requirements gathering methods, including interviews, workshops, surveys, and document analysis. The ability to effectively translate user needs into technical requirements.
- KU2.** Excellent communication, interpersonal, and collaboration skills to work effectively with cross-functional teams (designers, engineers, manufacturing, marketing).
- KU3.** Ability to identify, engage, and manage stakeholder expectations effectively.
- KU4.** Understanding of project management methodologies (Agile, Waterfall) and their application in the software development process.
- KU5.** Ability to create clear, concise, and comprehensive documentation (requirements specifications, design documents, API documentation).
- KU6.** Ability to identify and assess potential risks associated with the project, and develop mitigation strategies.
- KU7.** Understanding of relevant industry regulations and standards (e.g., data privacy, safety) and how they apply to the software development process.
- KU8.** Strong understanding of software development lifecycle (SDLC), requirements engineering, software design patterns (MVC, microservices), and object-oriented programming principles.
- KU9.** Proficiency in designing and implementing relational and NoSQL databases, including schema design, normalization, and query optimization. Familiarity with database management systems (DBMS) like MySQL, PostgreSQL, MongoDB, etc., is essential.
- KU10.** Knowledge in designing and implementing RESTful APIs, including understanding of HTTP methods (GET, POST, PUT, DELETE), API security, and API documentation. Familiarity with API testing tools is beneficial.
- KU11.** Knowledge of cloud platforms (AWS, Azure, GCP) and their services (IaaS, PaaS, SaaS), including cloud storage, serverless functions, and cloud security best practices. project costs
- KU12.** Basic understanding of networking concepts (TCP/IP, network security), and how applications interact with network infrastructure.
- KU13.** Familiarity with different operating systems (Windows, Linux, macOS) and their interaction with software applications.

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- KU14.** Basic understanding of hardware components (servers, storage, network equipment) and their role in supporting software applications.
- KU15.** Proficiency in at least one programming language relevant to the 3D printing domain. Python, C Plus Plus Java are common choice
- KU16.** While not strictly software, a fundamental understanding of 3D printing technologies (FDM, SLA, SLS, etc.) and their associated processes is crucial for effective requirements elicitation.
- KU17.** Proficiency in using Git or a similar version control system for managing code changes and collaborating with team members.
- KU18.** Knowledge of software testing methodologies (unit testing, integration testing, system testing), test automation frameworks, and bug tracking systems.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Undertake architecture related decisions on business and organization
- GS2.** follow rule-based decision-making processes
- GS3.** apply problem-solving approaches in different situations
- GS4.** provide opinions on work in a detailed and constructive way
- GS5.** work effectively in a customer facing environment

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Collaborate effectively with cross-functional teams (designers, engineers, manufacturing, marketing) to elicit and document comprehensive requirements. This includes understanding user needs, business goals, and technical constraints. Methods employed should include interviews, workshops, surveys, and document analysis.	2	3	-	1
PC2. Create a detailed requirements specification document covering both functional and non-functional requirements, this should include use cases, user stories, functional requirements and non-functional requirements	2	3	-	1
PC3. Create detailed description of the application's features and capabilities (e.g., slicing algorithms, printer control, material management, error handling, post-processing support, data visualization, reporting, information sharing, etc.).	2	4	-	1
PC4. Create specifications related to performance (speed, scalability, throughput), security (authentication, authorization, data protection), reliability, usability, maintainability, and compliance (regulatory standards).	2	4	-	2
PC5. Define the types and structures of data to be processed and stored by the application. This includes designing the database schema and considering data integrity, consistency, and security.	3	4	-	2
PC6. Identify and document all physical and virtual resources required for development, deployment, and operation of the 3D printing application. This includes hardware (servers, printers, network equipment), software (operating systems, databases, middleware), and personnel	3	4	-	2
PC7. Define the application infrastructure (architecture, deployment environment, network topology) necessary to support the 3D printing process. Address considerations such as high availability, scalability, and security	3	4	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. Determine how the application will integrate with existing systems (e.g., CAD software, ERP systems, cloud storage, other software applications). Detail the methods of integration (APIs, data exchange protocols, etc.) and address any interoperability challenges	3	4	-	2
PC9. Describe how the application will leverage other technologies such as AI/ML (for optimization and predictive maintenance), cloud computing (for scalability and cost-efficiency), IoT (for real-time monitoring and control), and other relevant technologies.	2	4	-	2
PC10. Identify and document the tools and processes needed for efficient infrastructure management (configuration management, monitoring, logging, security management, automated deployment).	2	4	-	2
PC11. Define security requirements and strategies to protect sensitive data and ensure compliance with relevant regulations (e.g., GDPR, HIPAA).	2	4	-	1
PC12. Define the scalability and capacity requirements of the application to ensure it can handle projected growth in data volume, user base, and processing demands. Document capacity planning methods used	2	4	-	1
PC13. Define appropriate SLAs for the 3D printing application, specifying performance targets, availability, and support	2	4	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8907
NOS Name	Determine the functional, non-functional, and infrastructural requirements for the defined scope of the software solution
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	3D Printing
NSQF Level	5
Credits	4
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8908: Translate the 3dp processes into relevant application storyboards, technical wireframes, and data flows in the software architecture

Description

This unit is about outlining the technical architecture and design patterns of the software applications in form of conceptual elaboration and intuitive visualization of the production processes

Scope

The scope covers the following :

- Storyboard development, Technical wireframing, Data flow diagramming, Architectural Design, Cloud Integration, Microservices Architecture, System Integration, GUI development

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Create comprehensive storyboards depicting the user flow and key interactions within the 3D printing application. These storyboards should illustrate different user scenarios and processes at a high level. The storyboard should reflect the overall user experience
- PC2.** Develop multiple versions of technical wireframes, focusing on key functionalities like slicing modules, printer management, material management, post processing support, data visualization and information sharing
- PC3.** Create detailed data flow diagrams illustrating the flow of information across the 3D printing system. This should encompass data from input (3D models, parameters), through processing (slicing, printer control), to output (printed object, status reports).
- PC4.** Design the overall software architecture for a cloud-based 3D printing application that integrates multiple systems
- PC5.** Detail the integration with cloud storage services (AWS S3, Azure Blob Storage, etc.) for storing and managing 3D models, print settings, and print results.
- PC6.** Compare a microservices-based architecture to a monolithic architecture, justifying the choice of architecture in the context of 3D printing, focusing on scalability, maintainability, and resilience. Identify specific services that would be suitable for decoupling (e.g., slicing, printer control, post-processing).
- PC7.** Document how the application integrates with external systems (e.g., CAD software, CAM software, material databases).
- PC8.** Develop a user-friendly graphical user interface (GUI) for interacting with the 3D printer's firmware. The GUI should allow users to control printer settings, monitor the print process, and handle potential errors. Wireframes and mockups should be created and thoroughly tested (usability testing is highly recommended).

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Excellent communication, interpersonal, and collaboration skills to effectively present design choices to stakeholders. The ability to receive and incorporate feedback.
- KU2.** Ability to identify and solve design problems creatively and efficiently. This includes evaluating different design approaches and making appropriate trade-offs.
- KU3.** Ability to clearly and effectively communicate design choices and rationale to stakeholders through storyboards, wireframes, data flow diagrams, and presentations.
- KU4.**
 - Ability to create clear, concise, and comprehensive document
 -
- KU5.** Ability to translate user needs and business requirements into effective design specifications.
- KU6.** Basic understanding of project management methodologies and their application to software design projects.
- KU7.** Understanding of software architecture principles and patterns (microservices, MVC, etc.). Ability to design scalable and maintainable architectures suitable for cloud deployment.
- KU8.** Proficiency in software design methodologies and techniques, including UML diagramming (use case diagrams, class diagrams, sequence diagrams, etc.).
- KU9.** Ability to create and interpret data flow diagrams (DFDs) to represent the flow of information within a system. Understanding of data structures and database design principles.
- KU10.** Strong understanding of cloud platforms (AWS, Azure, GCP) and their services (IaaS, PaaS, SaaS). Experience in designing and implementing cloud-based applications.
- KU11.** Experience in designing and implementing APIs (RESTful principles) for integrating with various systems.
- KU12.** Proficiency in designing and developing user interfaces (UI) using GUI frameworks (e.g., React, Angular, WPF, etc.). Understanding of UI/UX principles and best practices.
- KU13.** working knowledge of 3D printing processes (slicing, print preparation, post-processing) is essential to accurately represent the workflow in storyboards and data flow diagrams.
- KU14.** programming skills will be needed to implement some aspects of the design (e.g., creating prototypes of the GUI).

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Undertake architecture related decisions on business and organization
- GS2.** follow rule-based decision-making processes
- GS3.** apply problem-solving approaches in different situations
- GS4.** provide opinions on work in a detailed and constructive way
- GS5.** work effectively in a customer facing environment

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Create comprehensive storyboards depicting the user flow and key interactions within the 3D printing application. These storyboards should illustrate different user scenarios and processes at a high level. The storyboard should reflect the overall user experience	3	6	-	2
PC2. Develop multiple versions of technical wireframes, focusing on key functionalities like slicing modules, printer management, material management, post processing support, data visualization and information sharing	3	6	-	2
PC3. Create detailed data flow diagrams illustrating the flow of information across the 3D printing system. This should encompass data from input (3D models, parameters), through processing (slicing, printer control), to output (printed object, status reports).	3	6	-	2
PC4. Design the overall software architecture for a cloud-based 3D printing application that integrates multiple systems	3	6	-	2
PC5. Detail the integration with cloud storage services (AWS S3, Azure Blob Storage, etc.) for storing and managing 3D models, print settings, and print results.	4	6	-	3
PC6. Compare a microservices-based architecture to a monolithic architecture, justifying the choice of architecture in the context of 3D printing, focusing on scalability, maintainability, and resilience. Identify specific services that would be suitable for decoupling (e.g., slicing, printer control, post-processing).	4	6	-	3
PC7. Document how the application integrates with external systems (e.g., CAD software, CAM software, material databases).	5	7	-	3

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. Develop a user-friendly graphical user interface (GUI) for interacting with the 3D printer's firmware. The GUI should allow users to control printer settings, monitor the print process, and handle potential errors. Wireframes and mockups should be created and thoroughly tested (usability testing is highly recommended).	5	7	-	3
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8908
NOS Name	Translate the 3dp processes into relevant application storyboards, technical wireframes, and data flows in the software architecture
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	3D Printing
NSQF Level	5
Credits	4
Version	2.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8909: Build high-performance software applications by developing embedded code, data structures and algorithms

Description

This unit is about developing an ecosystem of scalable and interoperable, service-oriented software solutions for print-preparation and production processing

Scope

The scope covers the following :

- Parameter Optimization,Simulation,Firmware Development,Code Development,Data Integration,Software Integration,Deployment,File Import Enhancement,Security Implementation,Risk Management,Application Optimization,Testing and Quality Assurance

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Develop algorithms for optimizing printing parameters (e.g., layer height, extrusion speed, temperature) to improve print quality, speed, and material usage.
- PC2.** Create algorithms for simulating the 3D printing process to predict outcomes and identify potential problems before printing. This could include simulations of material flow, heat transfer, or structural integrity
- PC3.** Design and develop firmware to be installed on 3D printers. The firmware should efficiently control various aspects of the printer operation (Motion control, G-Code interpretation,Error handling)
- PC4.** Design, develop, and maintain efficient, reusable, and reliable code modules. Adhere to coding best practices (e.g., modular design, proper commenting, code reviews).
- PC5.** Develop pipelines to extract and process data from multiple sources like databases, knowledge bases
- PC6.** Leverage API connections and scripts to seamlessly integrate with existing software applications and packages (e.g., CAD software, cloud storage services).
- PC7.** Develop and deploy software applications to both on-premise and cloud environments. Address different deployment strategies and technologies (e.g., containerization, serverless functions).
- PC8.** Identify methods to enhance file import capabilities, supporting a broader range of file formats (STL, OBJ, etc.) and handling large files efficiently.
- PC9.** Implement security measures to protect application data and prevent unauthorized access. This might include data encryption, network security, authentication and Authorization
- PC10.** Identify potential risks and threats (e.g., cybersecurity vulnerabilities, buffer overflow, data corruption). Develop comprehensive mitigation plans to address these risks.
- PC11.** Optimize the software application for performance, scalability, and resource utilization. Incorporate techniques such as code optimization, database tuning, and caching. Address industry requirements for speed and efficiency

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PC12. Implement a comprehensive testing strategy that includes unit testing, integration testing, automated testing

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Ability to work effectively within a team, sharing knowledge and collaborating on software development tasks. Effective communication is crucial.
- KU2.** Understanding of project management principles and the ability to manage time, resources, and tasks effectively.
- KU3.** Ability to identify and solve complex problems related to software development, testing, and deployment.
- KU4.** Ability to identify and assess potential risks in the development process and develop mitigation strategies.
- KU5.** Proficiency in designing, implementing, and analyzing algorithms for performance optimization, especially in the context of 3D printing processes (slicing, path planning, parameter optimization). Understanding of algorithmic complexity (Big O notation) is highly beneficial.
- KU6.** Knowledge in developing and debugging firmware for embedded systems, specifically for 3D printers. Familiarity with microcontroller architectures, low-level programming, and real-time operating systems (RTOS). Knowledge of G-code and its interpretation is crucial.
- KU7.** Proficiency in various software development methodologies (Agile, Waterfall) and their practical application. Understanding of software design patterns (e.g., MVC, Singleton) and best coding practices.
- KU8.** Strong programming skills in languages commonly used for embedded systems (C, C PLUS PLUS) and potentially scripting languages (Python) for data processing and API integration.
- KU9.** Experience in designing and working with databases, extracting data, and ensuring data integrity. Understanding of SQL or NoSQL databases.
- KU10.** Skills in integrating software applications using APIs (REST, SOAP, etc.). Experience with API documentation and testing tools (e.g., Postman).
- KU11.** Understanding of cloud platforms (AWS, Azure, GCP), containerization technologies (Docker, Kubernetes), and serverless functions. Experience in deploying and managing applications in cloud environments.
- KU12.** Understanding of fundamental data structures (arrays, linked lists, trees, graphs) and their applications in efficient algorithm design.
- KU13.** Understanding of common security threats and vulnerabilities (e.g., buffer overflow, SQL injection). Proficiency in implementing security measures such as data encryption, authentication, and authorization.
- KU14.** Experience in various software testing methodologies (unit testing, integration testing, system testing). Familiarity with test automation frameworks (e.g., pytest, JUnit) and the creation of automated test cases.

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** Undertake architecture related decisions on business and organization
- GS2.** follow rule-based decision-making processes
- GS3.** apply problem-solving approaches in different situations
- GS4.** provide opinions on work in a detailed and constructive way
- GS5.** work effectively in a customer facing environment

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Develop algorithms for optimizing printing parameters (e.g., layer height, extrusion speed, temperature) to improve print quality, speed, and material usage.	2	3	-	1
PC2. Create algorithms for simulating the 3D printing process to predict outcomes and identify potential problems before printing. This could include simulations of material flow, heat transfer, or structural integrity	2	3	-	1
PC3. Design and develop firmware to be installed on 3D printers. The firmware should efficiently control various aspects of the printer operation (Motion control, G-Code interpretation, Error handling)	2	3	-	1
PC4. Design, develop, and maintain efficient, reusable, and reliable code modules. Adhere to coding best practices (e.g., modular design, proper commenting, code reviews).	2	3	-	1
PC5. Develop pipelines to extract and process data from multiple sources like databases, knowledge bases	2	4	-	2
PC6. Leverage API connections and scripts to seamlessly integrate with existing software applications and packages (e.g., CAD software, cloud storage services).	2	4	-	2
PC7. Develop and deploy software applications to both on-premise and cloud environments. Address different deployment strategies and technologies (e.g., containerization, serverless functions).	3	5	-	2
PC8. Identify methods to enhance file import capabilities, supporting a broader range of file formats (STL, OBJ, etc.) and handling large files efficiently.	3	5	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC9. Implement security measures to protect application data and prevent unauthorized access. This might include data encryption, network security, authentication and Authorization	3	5	-	2
PC10. Identify potential risks and threats (e.g., cybersecurity vulnerabilities, buffer overflow, data corruption). Develop comprehensive mitigation plans to address these risks.	3	5	-	2
PC11. Optimize the software application for performance, scalability, and resource utilization. Incorporate techniques such as code optimization, database tuning, and caching. Address industry requirements for speed and efficiency	3	5	-	2
PC12. Implement a comprehensive testing strategy that includes unit testing, integration testing, automated testing	3	5	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8909
NOS Name	Build high-performance software applications by developing embedded code, data structures and algorithms
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	3D Printing
NSQF Level	5
Credits	5
Version	2.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8910: Monitor, maintain and manage applications throughout the software development lifecycle

Description

This unit is about monitoring, managing and maintaining the applications throughout the software development lifecycle

Scope

The scope covers the following :

- Monitor Software application, Documentation, Feedback, version control

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Analyze and record the 3dp process variability in the output
- PC2.** Perform RCA to identify bottlenecks across the 3dp application and brainstorm solutions to these problems
- PC3.** Ensure the software is updated on a regular basis with the latest versions
- PC4.** Develop capabilities for automation and customization
- PC5.** Share application performance reports with relevant stakeholders
- PC6.** Prepare documentation of all software updates and bug fixes for internal reference
- PC7.** Collaborate with various teams, product partners and clients to receive constructive feedback on how to enhance the software application use
- PC8.** Maintain a backup of code base before proceeding to make any changes in the application

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures and guidelines which relate to evaluating and developing new use cases
- KU2.** different data sources and how to access documents and information from data sources
- KU3.** organizational policies and procedures for documenting new developed business cases
- KU4.** the range of standard templates and tools available and how to use them
- KU5.** how to evaluate business processes
- KU6.** how to evaluate new technologies and use cases
- KU7.** how to prepare reports
- KU8.** how to fix bugs and and update software
- KU9.** how to operate source control
- KU10.** Knowledge of 3D printing software lifecycle
- KU11.** Knowledge of 3D printing automation and customization tools

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Undertake architecture related decisions on business and organization
- GS2.** Analyze business impact and disseminate relevant information to others
- GS3.** Analyze data and understand its implications on business
- GS4.** Conduct impact analysis of the various actions performed and disseminating relevant information to others
- GS5.** follow rule-based decision-making processes
- GS6.** make decisions on suitable courses
- GS7.** apply problem-solving approaches in different situations
- GS8.** provide opinions on work in a detailed and constructive way
- GS9.** work effectively in a customer facing environment

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Analyze and record the 3dp process variability in the output	3	6	-	2
PC2. Perform RCA to identify bottlenecks across the 3dp application and brainstorm solutions to these problems	3	6	-	2
PC3. Ensure the software is updated on a regular basis with the latest versions	3	6	-	2
PC4. Develop capabilities for automation and customization	3	6	-	2
PC5. Share application performance reports with relevant stakeholders	4	6	-	3
PC6. Prepare documentation of all software updates and bug fixes for internal reference	4	6	-	3
PC7. Collaborate with various teams, product partners and clients to receive constructive feedback on how to enhance the software application use	5	7	-	3
PC8. Maintain a backup of code base before proceeding to make any changes in the application	5	7	-	3
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8910
NOS Name	Monitor, maintain and manage applications throughout the software development lifecycle
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	3D Printing
NSQF Level	5
Credits	3
Version	2.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment will consist of a blend of hands-on practical evaluations, viva-voce, and online proctored scenario-based multiple-choice questions ensuring a thorough evaluation of the individual's proficiency in learning outcomes, practical understanding, and real-world application of concepts.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8907.Determine the functional, non-functional, and infrastructural requirements for the defined scope of the software solution	30	50	-	20	100	21
SSC/N8908.Translate the 3dp processes into relevant application storyboards, technical wireframes, and data flows in the software architecture	30	50	-	20	100	21
SSC/N8909.Build high-performance software applications by developing embedded code, data structures and algorithms	30	50	-	20	100	21
SSC/N8910.Monitor, maintain and manage applications throughout the software development lifecycle	30	50	-	20	100	21
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	16
Total	140	230	-	80	450	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
IT-ITeS	Information Technology - Information Technology enabled Services
BPM	Business Process Management
BPO	Business Process Outsourcing
KPO	Knowledge Process Outsourcing
LPO	LPO
IPO	Information Process Outsourcing
IT-ITeS	Information Technology - Information Technology enabled Services
BPM	Business Process Management
BPO	Business Process Outsourcing
KPO	Knowledge Process Outsourcing
LPO	LPO
IPO	Information Process Outsourcing

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Helpdesk	Helpdesk is an entity to which the customers will report their IT problems. IT Service Helpdesk Attendant is responsible for managing the helpdesk.
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